



Government of India
Ministry of Agriculture and Farmers Welfare
Department of Agriculture and Farmers Welfare



Soil Health Card

Soil Health Card

API Integration Guidelines

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1.Summary

This document acts as a supporting document with the objective of assisting organisations for rapid and effective adoption of the Soil Health Card APIs to their benefit. It provides guidelines on how to design, implement, and consume APIs based on Soil Health Card portal and Mobile Application. Its intended audience includes technical architects, solution architects, software development teams, hosting and operations teams, and in general any entity that needs to set up and interact with the Soil Health Card portal. The policy is available at <https://soilhealth.dac.gov.in/privacy-policy>

The document will evolve over time, incorporating the experience gained from several implementations of the policy. The reader is therefore advised to refer to the latest available version for up-to-date content.

2.Introduction

The Ministry of Agriculture and Farmers' Welfare introduced the Soil Health Card (SHC) scheme on 19th February 2015. A soil health card provides information on the nutrient status of soil, along with recommendations on the dosage of nutrients to be utilised for improving its fertility and health. SHC is a printed report which contains nutrient status of soil with respect to 12 nutrients: pH, Electrical Conductivity (EC), Organic Carbon (OC), Nitrogen (N), Phosphorus (P), Potassium (K), Sulphur (S), Zinc (Zn), Boron (B), Iron (Fe), Manganese (Mn), Copper (Cu) of farm holdings. SHC will be provided to all farmers in the country at an interval of 3 years to enable the farmers to apply recommended doses of nutrients based on soil test values to realise improved and sustainable soil health and fertility, low costs and higher profits. Farmers can take a printout of the card from the SHC portal, which has a database of all harvesting seasons.

Soil Health Card portal

Soil Health Card portal is a Web based Workflow Application to generate Soil Health Cards in a uniform format throughout the country. It provides provision for Soil Sample Registration, Test Results entry, Automatic calculations of the fertilisers recommendations based on General Fertilisers Recommendations provided by States and finally, generation of Soil Health Card in local languages.

Soil Health Card Application consists of five major modules:-

- User Registration and its approval by State Administrator

- Sample Registration
- Test Result Entry by Analyst
- Fertiliser Recommendation
- Soil Health Card Generation

This document gives detailed instructions regarding the operations of the Soil Health Card portal.

SHC Web portal is available at URL: <https://soilhealth.dac.gov.in/>. The soil health card portal provides all the information needed about the Soil Health Card Scheme. Following features are available in the SHC Portal:

- Soil Maps with state-wise nutrient details
- Scheme progress and progress of portal entry
- Login to Admin Panel
- Farmers corner to Track soil sample, print soil health card, fertiliser dosage for crop, and locate soil testing laboratory
- Technical documentations, Guidelines & Downloads
- Directories & E-learning
- Contact details

SHC Admin Panel:

The salient features of the SHC admin panel are:

- Create Authorities like state, district authorities
- Soil testing labs run by Village Level entrepreneurs (VLEs)
- Every level of authority is managed by users. Users are assigned with a list of permissions to manage their roles.

- Each authority gets assigned with a number of allotted tests to be completed before the expiry date. After the completion of the tests, the authority gets payment which are then paid to the low-level authorities and Soil Testing Labs (STLs)
- The Vendors and device manufacturer's empanelment for inclusion of their services related to Soil Health Card.
- The Village Level Entrepreneurs (VLEs) need to use the soil sample collection App to perform the soil tests under the SHC portal.

The following sections give detailed overview and operational guidelines of the admin panel.

SHC Sample Collection Mobile App

SHC Sample Mobile collection is used by Village Level Entrepreneurs (VLEs) & Soil Testing Laboratory (STL) staff. Salient features of SHC Mobile App are:

- Farmer Registration by VLEs
- Sample Collection & Tracking by VLEs
- Conduct Soil test by STL staff
- Soil test results entry by Government labs
- Claim Submission, FAQs etc
- The App is available in English, Hindi & other regional languages

3. Purpose of document

- Used as a guideline by various Government of India departments to develop, publish and implement the APIs for their e-Governance Systems (services, e-Governance applications and systems)
- To help quick implementation and transparent monitoring of APIs and its usage.

4. Definitions

SHC:

Soil Health Card Scheme

STL:

Soil Testing Laboratory

State User:

State-level officer who will monitor and administer Soil Health Card scheme in a state.

District User:

District-level officer who will monitor and administer Soil Health Card scheme in a district

Agents:

Field Workers or Block-level officers or AEOs whose responsibilities include collecting soil samples

API Consumer :

An API consumer might be another organisation or government agency or an Indian organisation or other organisation with a registered application for use of an API. API consumers are free to use the published APIs in compliance with the terms of use subject to authentication and authorisation under which the API use is permitted. API consumers can rely on the published API definition without worrying about the specific implementation details (including implementation platform, internal architecture, etc.). API Consumer has the responsibility to strictly adhere to the terms of service as defined in these API guidelines.

QR Code:

Quick Response code used to identify the soil sample collected from the farmers. QR code must be pasted on to the Soil sample pouch. Sample will be registered using the mobile app by scanning the QR code. The QR code will be scanned in the mobile app by STL users for entering the test results.

5.Skillsets & Tools Required

- GraphQL & JWT (Json Web Token)
- Postman or Firecamp API Testing tool

6.API Registration

API Consumers that join SHC need to register their users and applications on the SHC Portal. API consumers will be able to access APIs only after registration. After successful completion of the API consumer on the SHC portal, API tokens will be mailed to access the SHC APIs. To register in the SHC portal, you need to send an email to soilhealthcard.support@krishitantra.com

7.API Integration Architecture

API Integration Architecture

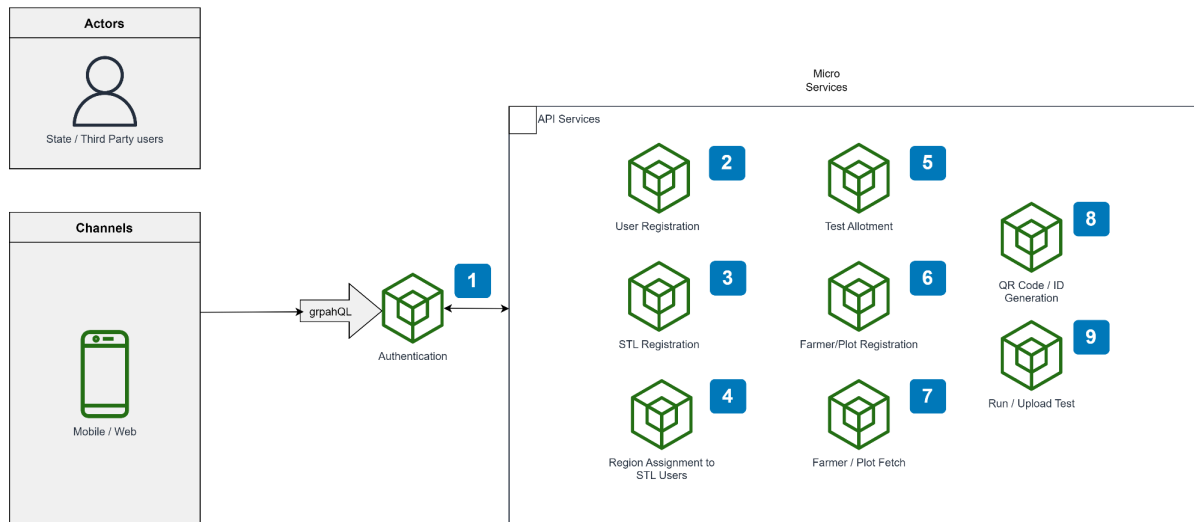


Fig. API Integration Architecture

- 1 API Consumers that join SHC need to register their users and applications on the SHC Portal. API consumers will be able to access APIs only after registration. After successful completion of the API consumer on the SHC portal, API tokens will be mailed to access the SHC APIs.
- 2 State-level & district-level officials can be registered on the SHC portal by using the user registration. The registrations of district user must be approved by the state user. Till the approval is done, the district user account will be displayed as Pending for Approval.
- 3 The details of the STL like name, address, location, phone number, etc. must be entered on the SHC portal and an account must be created for the STL user. After the STL registration, the STL will be listed on the SHC portal and tests can be allotted by the district user to the STL. Agents or Field workers will be created by STL users for the farmer registration and Soil sample collection using SHC mobile App.
- 4 Regions for STL can be allotted from where the STLs are allowed to collect soil samples. Two or three districts may use one STL for testing the soil samples in few states. Mapping the region for the STL is very essential.
- 5 Tests are allotted to States from Central admin, to districts from state admin and then to STLs from district admin. The STLs can collect soil samples only after the tests are allotted to the STLs by the district user.
- 6 The details of farmer like name, address, phone number etc. need to be entered to register the farmer. Once the farmer is registered soil samples can be collected from his plot for testing. Once the test results are entered, the farmer can download the soil health card from the portal.
- 7 After the farmer's registration, the plot details like location, soil color, crops grown, fertilizers used etc. are entered to register the plot. Image of the plot and location of the plot is needed. A QR code is used for sample collection. An unique sample code is generated during plot registration and this must be used to submit the soil test results.
- 8 QR codes are used to identify the soil samples. QR codes can be generated by STL officials and then need to be distributed to agents/ field workers. The QR code must be securely attached on the soil collection pouch and need to be scanned during Plot registration. In the labs, the QR code is again scanned before entering the soil test results.
- 9 The test results of 12 soil parameters can be uploaded to the SHC portal to generate the Soil Health Card. An unique sample code must be passed along with the soil test results.

8.API Protocol

I. Authentication

Authenticate the user with the system and obtain the auth_token

Request

GraphQL query for generate the Token

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in

Type	Params	Values
HEAD	token	string

Token

Authorization: bearer token must be sent with all client request. The token helps the server to validate the request source.

Type	Params	Input
Query	<pre>query Query(\$refreshToken: String!) { generateAccessToken(refre sh Token: \$refreshToken) }</pre>	<pre>{ "refreshToken": "refreshToken" }</pre>

Response

Field	Response
-------	----------

token	{
refreshToken	
	"data": {
	"generateAccessToken": {
	"token": " ",
	"refreshToken": " "
	}
	}
	}

I. Get Scheme Progress:

The details of the scheme progress like target, number of samples collected, number of tests completed, number of farmers registered can be fetched. This query does not need an access token.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in

Type	Params	Input
Query	<pre>query Query(\$state: String, \$district: String, \$cycle: String) { getProgressReport(state: \$state, district: \$district, cycle: \$cycle) }</pre>	<pre>state: \$state, district: \$district, cycle: \$cycle</pre>

Note: 1. To get the data for all the states:"cycle": "2023-24"

2. to get data of a particular state, pass state ID.

```
{
  "cycle": "2023-24",
```

```

"state": "63f5ceb98cec41e6c95ce3a8"
}
3. to get data of particular district, pass district ID.
{
  "cycle": "2023-24",
  "district": "63f5cedc8cec41e6c95ce65a"
}

```

Response

Format	Response
Json	Response will be an object containing the scheme progress. Example: <pre> { "id": "63f5ceb98cec41e6c95ce3a8", "name": "DELHI", "allotment": 10000, "SampleCollected": 12, "TestCompleted": 5, "Farmers": 14 } </pre>

II. User Registration

State-level & district-level officials can be registered on the SHC portal by using the user registration.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
------	--------	-------

mutation	<pre> mutation CreateUser(\$user: UserInfo) { CreateUser(user: \$user) { id username department_name name email phone staffid region permissions status admin address usertype createdAt updatedAt } }</pre>	<pre> { "user": { "usertype": String, "username": String, "region": JSON, "phone": String, "name": String, "email": String } }</pre>
----------	--	---

JSON data

Parameter	Format
region	<pre> region: { state: "state id", district: "district id", village: "village id" }</pre>

Response

Format	Response
Json	Response will be an object containing the created users details.

User Details	
Usertype	String
Username	String
Region	JSON

Phone	String
name	String
email	String

III. User Login

State-level/ district-level and STL officials can logon to the portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
mutation	mutation Login(\$password: String!, \$usertype: String!) { login(password: \$password, usertype: \$usertype) }	{ "password":String!, "usertype": String!, "username": String, }

Response

Format	Response
Json	User can log in .

IV. Forget Password

Link to the registered-email will be sent in case the user forgot their login credentials.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
mutation	<pre>mutation ForgotPassword(\$username: String, \$email: EmailAddress) { ForgotPassword(username: \$username, email: \$email) }</pre>	<pre>{ "username": String, "email": String }</pre>

Response

Format	Response
Json	Response will be true or false .

V. Get User Details:

The details of the user like address, Phone number, email etc can be fetched from the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
------	--------	-------

Query	<pre> query GetUser(\$id: String, \$username: String) { getUser(_id: \$id, username: \$username) { usertype username updatedAt status region phone createdAt name staffid permissions id email department_name admin address } } </pre>	<pre> { "id": string, "username": String } </pre>
-------	--	---

Response

Format	Response
Json	Response will be an object containing the list of all the user details associated with the selected id and user name.

Get User Details	
Id	String
User Name	String

VI. Get Users Details:

The details of the user like address, Phone number, email etc can be fetched from the SHC portal.

Request

Method	URL (Development)
--------	-------------------

POST	https://shc-staging.ddns.net
-------------	---

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
Query	<pre> query GetUsers(\$authority: String, \$region: JSON, \$admin: Boolean, \$username: String, \$phone: PhoneNumber) { getUsers(authority: \$authority, region: \$region, admin: \$admin, username: \$username, phone: \$phone) { id username department_name name email phone staffid permissions status usertype } } </pre>	<pre> { "authority": String, "region": JSON, "admin": Boolean, "username": String, "phone": String } </pre>

JSON data

Parameter	Format
region	<pre> region: { state: "state id", district: "district id", village: "village id" } </pre>

Response

Format	Response
--------	----------

Json	Response will be an object containing the list of all the user details associated with the selected authority,region,admin,username and phone.
------	--

Get Users Details	
Authority	String
Region	JSON
Admin	Boolean
User Name	String
Phone	String

VII. Edit User Details:

The details of the user like address, Phone number, email etc can be modified from the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
------	--------	-------

mutation	<pre> mutation UpdateUserProfile(\$userid: String!, \$userInput: UpdateuserInput) { UpdateUserProfile(userid: \$userid, userInput: \$userInput) { address department_name email name staffid phone username } }</pre>	<pre> { "userid": String!, "userInput": { "username": String!, "phone": String!, "name": String!, "email":String, "department_name": String!, "address": String!, "staffid": String! } }</pre>
----------	---	--

Response

Format	Response
Json	Response will be an object containing the list of all the user details associated with the selected id and user name.

Edit User Details:	
User Id	String!
User Name	String!
Phone	String!
Name	String!
Email	String
Department Name	String!
Address	String!
Staff Id	String!

VIII. Approve User:

After registration of the district-level users, the state-level user must approve the user by verifying the details of the applicant.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
mutation	<pre>mutation EnableUser(\$userid: String!) { EnableUser(userid: \$userid) }</pre>	<pre>{ "userid": String! }</pre>

Response

Format	Response
Json	Response will be an object containing the list of all the approved user details associated with the selected user id.

Approve User Details:	
User Id	String!

IX. Disable User:

The user account can be disabled to prevent the user from accessing the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
mutation	mutation DisableUser(\$userid: String!) { DisableUser(userid: \$userid) }	{ "userid": String! }

Response

Format	Response
Json	Response will be an object containing the list of all the disable user details associated with the selected user id.

Disable User Details:	
User Id	String!

X. Delete User:

The user account can be deleted from the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
mutation	mutation DeleteUser(\$deleteUserId: String) { deleteUser(id: \$deleteUserId) }	{ "deleteUserId": String }

Response

Format	Response
Json	Response will be an object containing the list of all the delete user details associated with the selected delete user id.

Delete User Details:	
Delete User Id	String

XI. Get STL Details:

The details of the STL like name, address, location, phone number, etc can be fetched from the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
Query	query GetSTL(\$getStlId: String!) { getSTL(id: \$getStlId) { id name email phone address state district region locale logo governmentlab } }	{ "getStlId": String! }

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
query	query GetSTLs(\$region: JSON, \$statecode: String, \$districtcode: String, \$blockcode: String, \$username: String, \$level: String, \$getStLsId: String, \$status: String) { getSTLs(region: \$region, statecode: \$statecode, districtcode: \$districtcode, blockcode: \$blockcode, username: \$username, level: \$level, id: \$getStLsId, status: \$status) { id name phone email region details address username testcenter { id name phone address email	<pre>{ "region": { "state": String, "district": String }, "statecode": String, "districtcode": String, "blockcode": String, "username": String, "level": String, "getStLsId": String, "status": String }</pre>

	<pre> } Manufacturer { id name computedID email phone address appName appPackage link type static } createdAt updatedAt } } </pre>	
--	---	--

JSON data

Parameter	Format
region	<pre> region: { state: "state id", district: "district id", village: "village id" } </pre>

Response

Format	Response
Json	Response will be an object containing the list of all the STL user details associated with the selected getstlsid, user, region, government lab created by and level .

STLs Details:	
Get Stls Id	String
User	String
Region	JSON
User Name	String
Government Lab	Boolean
Created By	String
Level	String

XIII. Get STLs Details (Open API):

The details of the STL like name, address, location, phone number, etc can be fetched from the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
------	--------	-------

query	<pre> query GetTestCenters(\$state: String, \$district: String, \$statecode: String, \$districtcode: String) { getTestCenters(state: \$state, district: \$district, statecode: \$statecode, districtcode: \$districtcode) { id name phone email address region createdAt STLdetails { name email phone status details } } } </pre>	<pre> { "state": String, "district": String, "statecode": String, "districtcode": String } </pre>
-------	--	---

Response

Format	Response
--------	----------

Json	Response will be an object containing the list of all the STL details associated with the selected state and district .
------	---

TestCenters Details:	
state	String
district	String

XIV. Get Labs For School

This method is for getting the list of name,createdBy,type,manufacturer,state,district,village,address etc can be fetched from the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Values
HEAD	token	string

Type	Params	Input
------	--------	-------

Query	<pre> query GetLabsForSchool { getLabsForSchool { id name createdBy type Manufacturer { name } region createdAt updatedAt } }</pre>	
-------	---	--

Response

Format	Response
Json	Response will be an object containing the list of all name,createdBy,type,manufacturer,state,district,village,address data.

XV. Register STLs:

The details of the STL like name, address, location, phone number, etc can be fetched from the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
------	--------	-------

mutation	<pre> mutation CreateSTL(\$stlinput: STLInputs) { CreateSTL(stlinput: \$stlinput) { id name email phone address state district region locale logo governmentlab static username Bank { id accountnumber IFSC name PAN } testcenter { id name state district region timing phone email address authenticatedUser createdAt updatedAt } status details Manufacturer { id name type } village </pre>	<pre> { "stlinput": { "username": String!, "testcenter": { "region": JSON!, "phone": String!, "name": String!, "address": String!, "timing": JSON! }, "static": Boolean!, "region": JSON!, "phone": PhoneNumber, "name": String!, "email": String!, "details": JSON, "address": String } } </pre>
----------	---	---

	subdistrict createdAt updatedAt usertype } }	
--	---	--

JSON data

Parameter	Format
region	region: { state: "state id", district: "district id", village: "village id" }

Response

Format	Response
Json	Response will be an object containing the list of all the Register STL user details associated with the selected user inputs.

Register STL Details:

User Name	String
Region	JSON!
Phone	String
Name	String!
Address	String
Timing	JSON!

Static	Boolean!
Region	JSON!
Name	String!
Email	String!
Details	JSON
Address	String

XVI. Disable STL:

STLs can be disabled to prevent the STL officials from accessing the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
mutation	<pre> mutation DisableSTL(\$disableStlId: String!, \$status: [STATUS]) { DisableSTL(id: \$disableStlId, status: \$status) { id } }</pre>	<pre> { "disableStlId": String!, "status": [STATUS] { "Pending": "Active": } }</pre>

--	--	--

Response

Format	Response
Json	Response will be an object containing the list of all the disable user details associated with the selected disable stl id and status.

Disable STL Details:	
Disable Stl Id	String!
Status	[STATUS]

XVII. Edit STL Details:

The details of the STL like name, address, location, phone number, etc can be edited from the portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
mutation	mutation UpdateSTL(\$updateStlId: String!, \$stlinput: STLUpdateInputs!) { UpdateSTL(id: \$updateStlId, stlinput: \$stlinput) { id name email phone	{ "updateStlId": String!, "stlinput": { "username": String!, "region": JSON!, "name": String!, "email":

	address state district locale logo governmentlab static username status details createdAt updatedAt region usertype } }	EmailAddress!, "details": JSON, "address": String! } }
--	--	--

JSON data

Parameter	Format
region	region: { state: "state id", district: "district id", village: "village id" }

Response

Format	Response
Json	Response will be an object containing the list of all the selected stl details associated with the selected update stl id and stl inputs.

Edit STL Details:	
Update Stl Id	String!
User Name	String!
Region	JSON!

Name	String!
Email	EmailAddress!
Details	JSON
Address	String!

XVIII. Get Agent Details:

Fetch the details of agents & Field workers like their name, phone number, email etc.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
Query	<pre> query GetAgent(\$getAgentId: String) { getAgent(id: \$getAgentId) { id state district village name aadharNumber qualification permissions phone email username status address details parentUser createdAt updatedAt </pre>	<pre> { "getAgentId": String } </pre>

	<pre>} }</pre>	
--	----------------	--

Response

Format	Response
Json	Response will be an object containing the list of all the selected agent details associated with the selected agent id.

Agent Details:	
Get Agent Id	String

XIX. Get Agents Details:

Fetch the details of agents & Field workers like their name, phone number, email etc.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
Query	<pre>query GetAgents(\$parentUser: String, \$state: String, \$district: String, \$username: String) { getAgents(parentUser: \$parentUser, state: \$state, district: \$district, username: \$username) { aadharNumber address createdAt district email id name } }</pre>	<pre>{ "parentUser": String, "state": String, "district": String, "username": String }</pre>

	parentUser phone qualification state updatedAt username village } }	
--	---	--

Response

Format	Response
Json	Response will be an object containing the list of all the selected agents details associated with the selected parent user,state,district,username.

Agents Details:	
Parent User	String
State	String
District	String
User Name	String

XX. Edit Agent Details:

Change the details of agents & Field workers like their name, phone number, email etc.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
mutation	<pre> mutation UpdateAgent(\$updateAgentId: String, \$agent: AgentInput) { UpdateAgent(id: \$updateAgentId, agent: \$agent) { id state district village name aadharnumber qualification permissions phone email username status address details parentUser createdAt updatedAt } }</pre>	<pre> { "updateAgentId": String, "agent": { "village": String, "username": String!, "state": String!, "phone": String!, "email": String, "name": String!, "address": String } }</pre>

Response

Format	Response
Json	Response will be an object containing the list of all the selected edit agent details associated with the selected update agent id and agent details .

Edit Agent Details:	
Update Agent Id	String
Village	String
User Name	String!
State	String!

Phone	String!
Email	String
Name	String!
Address	String

XXI. Get Test allotment Details:

Fetch the details of the Test allotment.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
Query	query GetallotmentCoupon(\$getallotmentCouponId: String!) { getallotmentCoupon(id: \$getallotmentCouponId) { name id belongsto { username phone name id email } allparameters allotedTestCount completedTestCount testparametersprice totalcount testparameters region } }	<pre>{ "getallotmentCouponId": String! }</pre>

	<pre> price Expire createdAt updatedAt } } </pre>	
--	---	--

Response

Format	Response
Json	Response will be an object containing the list of all the selected test allotment details associated with the selected get allotment coupon id.

Test Allotment Details:	
Get Allotment Coupon Id	String!

XXII. Get Test allotment Details:

Fetch the details of the Test allotment.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
Query	<pre> query GetallotmentCouponBybelongs(\$belongsto: String, \$region: String,\$createdby: String) { getallotmentCouponBybelongs(belongsto: \$belongsto, region: \$region,createdby:\$createdby) { </pre>	<pre> { "region": String, "createdby": String, "belongsto": String } </pre>

	<pre> name id belongsto { username phone name id email } allparameters allotedTestCount completedTestCount testparametersprice totalcount testparameters region price Expire createdAt updatedAt } } </pre>	
--	---	--

Response

Format	Response
Json	Response will be an object containing the list of all the selected test allotment details associated with the selected get allotment region,createdby and belongsto.

Test Allotment Details:	
Region	String
CreatedBy	String
BelongsTo	String

XXIII. Farmer Registration:

The details of farmer like name, address, phone number etc can be entered to register the farmer.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
mutation	<pre> mutation CreateFarmer(\$farmer: farmerInput) { CreateFarmer(farmer: \$farmer) { id belongsTo name locale address phone location email state district aadhaar details disabled createdAt updatedAt } }</pre>	<pre> { "farmer": { "village": String!, "state": String!, "phone": String, "name": String!, "location": JSON, "locale": String!, "email": String, "district": String!, "details": JSON, "address": String!, "aadhaar": String } }</pre>

JSON data

Parameter	Format
location	<pre> location:{ type:"Point", coordinates:[77.1841789,28.6676837] }</pre>
details	<pre> details: {</pre>

	sub_district_id: subdistrict id(String), }
--	---

Response

Format	Response
Json	Response will be an object containing the list of all the selected Farmer Registration details associated with the selected name,village,phone and address.

Farmer Registration Details:	
Village	String!
State	String!
Phone	String
Name	String!
Location	JSON
Locale	String!
Email	String
District	String!
Details	JSON
Address	String!
Aadhaar	String

XXIV. Plot Registration:

After the farmer's registration, the plot details like location, soil colour, crops grown, fertilisers used etc are entered to register the plot.

Request

Method	URL (Development)
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POST	https://shc-staging.ddns.net
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Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
mutation	<pre> mutation CreatePlot(\$farmer: ID!, \$plot: PlotInput!) { createPlot(farmer: \$farmer, plot: \$plot) { id location place mandal district state country pincode address soilType soilColor soilTexure soilDensity area resultUnits surveyNo stoniness landSloping drainage waterSource waterSourceCapacity irrigationMethod electricityConnectionPhase irrigationPumpSize dTconnection plotReferenceCode createdAt updatedAt image } }</pre>	<pre> { "farmer":String!, "plot": PlotInput! }</pre>

Response

Format	Response
Json	Response will be an object containing the list of all the selected Plot Registration details associated with the selected farmer and plot.

Plot Registration Details:	
Farmer	String!
Plot	PlotInput!

XXV. Bulk Upload

The details of the STL like name, address, location, phone number, etc can be fetched from the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://shcstaging.infinium.management

Type	Params	Input
Query	mutation CreateBulkTest(\$farmer: farmerInputForBulk!, \$plots: PlotInputForBulk!, \$test: testInputForBulk!) { createBulkTest(farmer: \$farmer, plots: \$plots, test: \$test) { id computedID QrcodeID {	{ "farmer": { name: String! locale: String address: String! phone: PhoneNumber location: JSON

	<pre> QrcodeID } createdBy farmer { id name phone } crop location html testparameters results status testCompletedAt sampleDate createdAt updatedAt state district village details printouts plot { id } } } </pre>	<pre> villagecode: String! email: EmailAddress aadhaar: String details: JSON } , "location": { "type": "Point", "coordinates": [74.8412769, 74.8412769] } } "plots": { mandal: String! address: String! country: "India", villagecode: String! soilType: String soilColor: String! soilTexure: String! soilDensity: Float area: Float! surveyNo: String resultUnits: "Hectare"! Image:string! (in Base64 format) , location: { "type": "Point", "coordinates": [74.8412769, 74.8412769] }! } "test":{ crop: String (upto 6 crops separated by comma) testCentre: String! results:{n : Float, p : Float, k : Float, B : Float, Fe : Float, Zn : Float, Cu : Float,S : Float, OC : Float, </pre>
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		pH : Float,EC : Float, Mn : Float }! testCompletedAt:Datetime sampleDate:Datetime }
--	--	--

Response

Format	Response
Json	Response will be an object containing the list of all the parameters computed ID, QRCode ID, farmer details, sample details, test details.

XXVI. Fetch Farmer Details:

The details of farmers like name, address, phone number etc are fetched from the database.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
Query	query GetFarmers(\$belongsTo: String, \$name: String, \$phone: String, \$state: String, \$district: String, \$village: String, \$statecode: String, \$districtcode: String, \$blockcode: String) { getFarmers(belongsTo: \$belongsTo, name: \$name, phone: \$phone, state: \$state, district: \$district, village: \$village, statecode: \$statecode,	{ "belongsTo": String, "name": String, "phone": String, "state": String, "district": String,

	districtcode: \$districtcode, blockcode: \$blockcode) { id belongsTo name locale address phone location email details disabled createdAt updatedAt } }	"village": String, "statecode": String, "districtcode": String, "blockcode": String }
--	--	---

Response

Format	Response
Json	Response will be an object containing the list of all the selected Fetch Farmer details associated with name,phone,state and district.

Fetch Farmer Details:	
Belongs To	String
Name	String
Phone	String
State	String
District	String

XXVII. Fetch Land Details:

The plot details like location, soil colour, crops grown, fertilisers used etc are fetched from the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
Query	<pre> query GetPlots(\$farmer: ID!) { getPlots(farmer: \$farmer) { waterSourceCapacity waterSource updatedAt surveyNo stoniness state soilType soilTexure soilDensity soilColor resultUnits mandal location district dTconnection drainage electricityConnectionPhase landSloping irrigationPumpSize irrigationMethod image createdAt } }</pre>	<pre> { "farmer": String! }</pre>

Response

Format	Response
Json	Response will be an object containing the list of all the selected Fetch Land details associated with farmer.

Fetch Land Details:	
Farmer	String!

XXVIII. Get Test Details:

The details of the test like sample number, farmer name, sample collection date, crops etc can be fetched from the SHC portal.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
Query	<pre> query GetTestDetails(\$from: String!, \$to: String!) { getTestDetails(from: \$from, to: \$to) { updatedAt testCentre { updatedAt timing state region phone name createdAt address } status sampleDate results recommendation plot { </pre>	<pre> { "from": String!, "to": String! }</pre>

	<pre> updatedAt stoniness state } farmer { phone name } createdAt computedID allottedCoupon { updatedAt totalcount } } } </pre>	
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Response

Format	Response
Json	Response will be an object containing the list of all the selected Test details associated with from and to .

Test Details:	
From	String!
To	String!

XXIX. Upload Test:

The test results of 12 soil parameters can be uploaded to the SHC portal to generate the Soil Health Card.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
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POST	https://soilhealth4.dac.gov.in/
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Type	Params	Input
mutation	<pre> mutation CreateTest(\$test: testInput!) { CreateTest(test: \$test) { id computedID device createdBy crop location html previouscrop previousyield previousyieldprice targetyield previousmanure previousfertilizerquantity testparameters results status testCompletedAt sampleDate createdAt updatedAt } }</pre>	<pre> { "test": { "testCentre": String!, "plot": String!, "previousyield": Int, "details": JSON, "crop": String, "allotedTest": Boolean!, "farmer": String!, "location": JSON } }</pre>

JSON data

Parameter	Format
region	<pre> region: { state: "state id", district: "district id", village: "village id" }</pre>
location	<pre> location: { type: "Point", coordinates: [77.1841789, 28.6676837] }</pre>

--	--

Response

Format	Response
Json	Response will be an object containing the list of all the selected upload Test details associated with test .

Upload Test Details:	
Test Centre	String!
Plot	String!
Previous Yield	Int
Details	JSON
Crop	String
Alloted Test	Boolean!
Farmer	String!
Location	JSON

XXX. Generate QR Codes:

QR codes are used to identify the soil samples. QR codes can be generated by STL officials and then need to be distributed to agents/ field workers.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
mutation	<pre> mutation CreateSampleCollection(\$sample: SampleCollectionInput) { CreateSampleCollection(sample: \$sample) { id QrcodeID name createdBy disabled createdAt updatedAt } }</pre>	<pre> { "sample": { "numberofsamplCol lections": Int!, "name": String, "disabled": Boolean } }</pre>

Response

Format	Response
Json	Response will be an object containing the generate qr code details associated with the sample.

Generate QR Code Details:	
Number Of Sample Collections	Int!
Name	String
Disabled	Boolean

XXXI. Fetch Soil Health Card

This method is for Generating the soil health card report based on state, district or Aadhar number and locale represents the language of the report.

Language Supported with locale code:

1. English(en) default
2. Hindi(hi)
3. Kannada(kn)

4. Telugu(te)
5. Tamil(tm)
6. Malayalam (ml)
7. Marathi(mr)
8. Punjabi(pa)
9. Bengali(ba)
- 10.Gujarathi(gu)
- 11.Odriya(or)

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Input
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Query	<pre> query GetTestForAuthUser(\$computedId: String, \$phone: PhoneNumber, \$state: String, \$district: String, \$name: String, \$farmer: String, \$from: Datetime, \$to: Datetime) { getTestForAuthUser(computedID: \$computedId, phone: \$phone, state: \$state, district: \$district, name: \$name, farmer: \$farmer, from: \$from, to: \$to) { id computedID plot { address area surveyNo } farmer { address name } crop location html(locale: \$locale) testparameters rdfValues status testCompletedAt sampleDate reportData district village results } } </pre>	<pre> { "state": String, "district": String, "name": String, "phone": String, "from": "yyyy-mm-dd", "to": "yyyy-mm-dd", "farmer": String, "computedId": String } </pre>
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Response

Format	Response
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Json	Response will be an object containing the list of all soil health card associated with the state, district or with an Aadhar number.
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XXXII. Get Fertiliser Recommendation for Crops:

This query will fetch the fertiliser recommendation for a crop based on NPK values, and state.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in

Type	Params	Input
Query	<pre>query Query(\$state: ID!, \$crops: [ID!], \$results: JSON!) { getRecommendations(state: \$state, crops: \$crops, results: \$results) }</pre>	<pre>\$state: ID!, \$crops: ID! \$results: JSON!</pre>

Response

Format	Response
Json	Response will be an object containing the crop name, region, nutrient requirement based on the range.

Test Details:	
state	ID!
crops	ID!
results	JSON!

XXXIII. Get Fertiliser Recommendation

This query will fetch the nutrient requirements for crops based on region.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in

Type	Params	Input
Query	<pre>query GetGFRs(\$state: ID, \$crop: String, \$district: String) { getGFRs(state: \$state, crop: \$crop, district: \$district) { crop { name id } region nutrients } }</pre>	<pre>\$state: String, \$crop: String \$district: String</pre>

Response

Format	Response
Json	Response will be an object containing the crop name, region, nutrient requirement based on the range.

Test Details:	
crop	JSON

region	JSON
nutrients	JSON

XXXIV. Get Crop

This query will fetch the crop names and corresponding ID based on state.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in

Type	Params	Input
Query	<pre> query GetCrops(\$state: String) { getCrops(state: \$state) { id name region GFRavailable } } </pre>	\$state: String,

Response

Format	Response
Json	Response will be an object containing the crop ID, crop name, region, and GFR availability status.

Test Details:	
ID	String
name	String

region	JSON
GFR Available	Yes/No

XXXV. Get School Progress

This query will fetch the scheme progress of School Soil health card program.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in

Type	Params	Input
Query	<pre>query Query(\$state: String) { getSchoolProgressReport(state: \$state) }</pre>	\$state: String,

Response

Format	Response
Json	Response will be an object containing the school ID, School name, type, address, state, district, number of batches, number of students, target Alloted, sample Collected, test Completed, farmers.

XXXVI. Fetch State

This method is for getting the state list with IDs.

Request

Method	URL (Development)
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POST	https://shc-staging.ddns.net
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Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Values
HEAD	token	string

Type	Params	Input
Query	query Query { getState }	

Response

Format	Response
Json	Response will be an object containing the list of all state names with IDs.

XXXVII. Fetch District and sub-district

This method is for getting the district and sub-district list with IDs.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Values
HEAD	token	string

Type	Params	Input
Query	<pre>query Query(\$state: ID, \$subdistrict: Boolean) { getdistrictAndSubdistrictBystate(state: \$state, subdistrict: \$subdistrict) }</pre>	<pre>{ "State":state id, "subdistrict": "true/false" }</pre>

Response

Format	Response
Json	Response will be an object containing the list of all districts if sub district is stated as false otherwise this query will return sub district list associated with the state id.

XXXVIII. Fetch village list based on district

This method is for getting the list of villages with IDs.

Request

Method	URL (Development)
POST	https://shc-staging.ddns.net

Method	URL (Production)
POST	https://soilhealth4.dac.gov.in/

Type	Params	Values
HEAD	token	string

Type	Params	Input
Query	<pre>query Query(\$district: ID!, \$subdistrict: String) { getVillageBydistrict(district: \$district, subdistrict: \$subdistrict) }</pre>	<pre>{ "district": district Id, "subdistrict": subdistrict id }</pre>

Response

Format	Response
Json	Response will be an object containing the list of all villages with Ids.

Revision History

Version	Date	Revisions
1.1	September 6, 2023	Added APIs: Fetch Soil Health Card and Fetch District based on state. Deleted API: CreateAllotment
1.2	October 10, 2023	Added APIs: getState, getVillageBydistrict
1.31	November 3, 2023	Updated CreateBulkTest for LGD codes
1.32	December 29, 2023	Added GetLabsForSchool
1.33	March 4, 2024	Updated dev URL Updated getTestForAuthUser with ComputedID and farmer id
1.34	March 6, 2024	Addition of Open API for Labs GetTestCenters
1.35	March 20, 2024	Addition of Open API for fetching lab details